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GitHub Link	https://github.com/192425292simats-byte/MLA0403-DEEP-LEARNING

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT : A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I (V Sunil Kumar / 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: V Sunil

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Total						

1. Article Details

Title: *DenseNet201Plus: Cost-effective transfer-learning architecture for rapid leaf disease identification with attention mechanisms*

Authors: Md. Khairul Alam Mazumder, Md. Mohsin Kabir, Ashifur Rahman, Md. Abdullah-Al-Jubair, and M.F. Mridha.

Journal: *Heliyon*, Elsevier

Year: 2024

Volume/Issue: Volume 10, Issue 15

Article Number: e35625

DOI: 10.1016/j.heliyon.2024.e35625

Topic: DenseNet, Transfer Learning, Attention Mechanism and Explainable AI

[View the research article](#)

2. Research Problem

Plant diseases can significantly reduce crop productivity, food security and farmers' income. Traditional disease identification often depends on **manual visual inspection**, which can be time-consuming, subjective and difficult when diseases have similar visual symptoms.

The researchers therefore aimed to develop an automated deep-learning system capable of identifying plant leaf diseases accurately, including images with **different leaf sizes and complicated backgrounds**.

Limitations of Existing Approaches

The article identifies several problems with existing approaches:

- Many existing models concentrate on a single plant or disease.
- Performance can decrease when images contain complicated backgrounds.
- Pre-trained models can suffer from underfitting or overfitting on challenging datasets.
- Some existing systems require considerable computational resources.
- Generalization to different plant species and real-world conditions remains challenging.

The researchers therefore proposed a modified DenseNet201 architecture to improve robustness and feature extraction.

3. Objectives of the Study

The major objectives are:

1. To develop an accurate automated plant leaf disease identification system.
2. To improve the performance of DenseNet201 through architectural modifications.

3. To use **transfer learning** to benefit from pre-trained CNN features.
4. To improve feature extraction using **dense blocks and attention mechanisms**.
5. To reduce the effects of overfitting and underfitting.
6. To evaluate the proposed model on two different plant disease datasets.
7. To improve model interpretability using **Grad-CAM++**.

The overall goal is to develop a model that is accurate, robust and useful for practical agricultural disease detection.

4. Dataset Description

The researchers used two publicly available datasets.

A. Banana Leaf Disease Dataset

It contains three classes:

- Healthy
- Segatoka
- Xanthomonas

Class	Training Testing	
Healthy	125	30
Segatoka	260	60
Xanthomonas	661	153
Total	1,046	243

Therefore, the Banana dataset contains **1,289 images**.

B. Black Gram Leaf Disease Dataset

It contains five classes:

- Anthracnose
- Healthy
- Leaf Crinkle
- Powdery Mildew
- Yellow Mosaic

Class	Training Testing	
Anthracnose	180	50
Healthy	179	42
Leaf Crinkle	122	30
Powdery Mildew	144	36

Class	Training Testing	
Yellow Mosaic	182	42
Total	807	200

Therefore, the Black Gram dataset contains **1,007 images**.

Overall Dataset

The study used:

$1,289 + 1,007 = 2,296$ images

The datasets were divided into approximately **80% training and 20% testing**. The study did not report a separate validation split in the dataset table; model-selection experiments were conducted using the training procedure.

Type of Data

The study uses **RGB leaf images**.

5. Proposed Methodology

The researchers proposed an improved version of DenseNet201 called **DenseNet201Plus**.

The methodology consists of:

Leaf Image → Preprocessing → Data Augmentation → DenseNet201 → Attention Mechanisms → Dense Blocks → Global Average Pooling → Dense Layer → Disease Prediction

Data Preprocessing

The images undergo several transformations, including:

- Rotation
- Shearing
- Height and width adjustments
- Horizontal flipping
- Pixel normalization

Pixel values originally ranging from **0 to 255** are normalized to the **0–1 range**. These techniques increase data diversity and help reduce overfitting.

Training

The researchers experimented with different:

- Input dimensions
- Batch sizes
- Epochs
- Data augmentation methods
- Optimizers

- Learning rates
- Activation functions

The experiments were conducted using **TensorFlow and Keras** on an **NVIDIA GeForce RTX 2060 Super GPU**.

6. Deep Learning Technique Used

DenseNet

DenseNet is a CNN architecture in which layers are densely connected.

Instead of:

Layer 1 → Layer 2 → Layer 3 → Layer 4

DenseNet uses connections such as:

Layer 1 → Layer 2, Layer 3, Layer 4

Layer 2 → Layer 3, Layer 4

Layer 3 → Layer 4

Therefore, later layers can directly access features learned by earlier layers.

Feature Reuse

For plant disease identification, early layers may learn:

- Edges
- Shapes
- Textures

Deeper layers can reuse these features to identify:

- Spots
- Lesions
- Color patterns
- Disease symptoms

This reduces unnecessary relearning and improves gradient flow.

Transfer Learning

The proposed approach starts with a **pre-trained DenseNet201**. Instead of learning all image features from the beginning, the model reuses knowledge obtained from a large image dataset and adapts it to plant disease classification.

This is particularly useful when the target dataset is relatively small.

Attention Mechanism

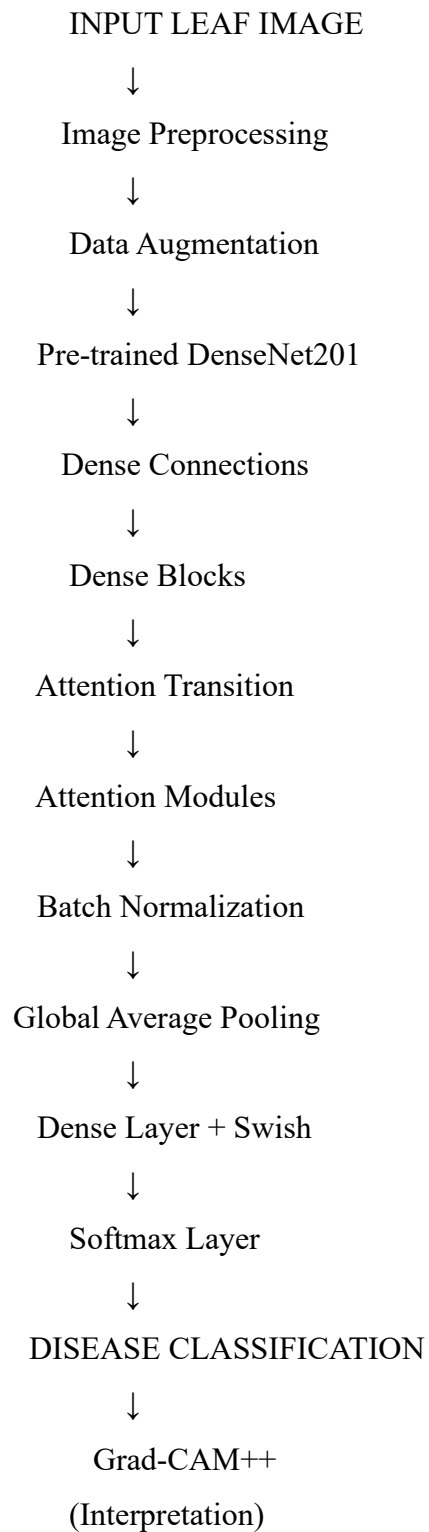
Attention mechanisms help the network concentrate on **important regions and features** associated with disease symptoms.

Grad-CAM++

Grad-CAM++ was incorporated to improve interpretability. It helps visualize the regions of an image that influenced the model's prediction, allowing researchers and agricultural practitioners to understand where the disease-related evidence is located.

7. Model Architecture

A simplified architecture of the proposed system is:



Role of Major Components

Input: Receives RGB plant leaf images.

Preprocessing: Normalizes and standardizes the images.

Data Augmentation: Creates variations of training images to improve generalization.

DenseNet201: Extracts hierarchical visual features.

Dense Blocks: Provide dense connections and enable feature reuse.

Attention Modules: Focus on important disease-related information.

Global Average Pooling: Converts feature maps into a compact representation.

Dense Layer: Learns high-level classification features.

Softmax: Produces the probability of each disease class.

Grad-CAM++: Provides visual explanation of the model's decision.

8. Performance Evaluation

The main metrics used were:

- Accuracy
- Precision
- Recall
- AUC
- Average training time per epoch
- GPU inference time

The proposed model achieved the following results.

Dataset	Accuracy	Precision	Recall	AUC
Banana Leaf Disease	90.12%	90.12%	90.12%	97.16%
Black Gram Leaf Disease	99.50%	99.50%	99.50%	100%

Comparison

For the Banana dataset:

- DenseNet201: 82.30%
- Modified DenseNet201 with PReLU: 89.71%
- Modified DenseNet201 with ReLU: 89.71%
- **Proposed DenseNet201Plus: 90.12%**

For the Black Gram dataset:

- DenseNet201: 91.50%
- Modified DenseNet201 with PReLU: 97.00%
- Modified DenseNet201 with ReLU: 99.00%
- **Proposed DenseNet201Plus: 99.50%**

9. Results and Discussion

The proposed DenseNet201Plus performed better than the evaluated pre-trained CNN models on both datasets.

Major Findings

1. The proposed architecture achieved **90.12% accuracy** on the Banana dataset.
2. It achieved **99.50% accuracy** on the Black Gram dataset.
3. The Black Gram dataset achieved an AUC of **1.0**.
4. DenseNet201Plus performed better than the original DenseNet201.
5. Attention mechanisms improved the model's ability to focus on important features.
6. The Swish activation function provided better performance than the tested PReLU and ReLU configurations in the final architecture.
7. Grad-CAM++ improved model interpretability.

The difference between the datasets is important. The Banana dataset contains more challenging background conditions, and the proposed model still achieved good performance. The Black Gram dataset produced extremely high performance, suggesting that its visual patterns were easier for the model to distinguish.

10. Advantages of the Proposed Model

At least three important advantages are:

1. High Classification Accuracy

The model achieved **90.12% accuracy for banana disease detection and 99.50% for black gram disease detection**.

2. Effective Feature Reuse

Dense connections allow earlier features to be reused by deeper layers, improving feature propagation and reducing redundant learning.

3. Transfer Learning

The use of a pre-trained DenseNet reduces the amount of data and training required compared with training a deep network completely from scratch.

4. Better Interpretability

Grad-CAM++ provides visual explanations of the regions responsible for predictions.

5. Robustness

The model was tested on datasets with different leaf sizes and background complexity.

6. Practical Agricultural Application

The approach could support automated plant disease identification and early intervention.

11. Limitations of the Study

Although the results are strong, several limitations should be considered.

1. Limited Dataset Size

The total number of images is only **2,296**, which is relatively small for a deep-learning model.

2. Dataset Diversity

Only two plant datasets were evaluated. Therefore, the model's ability to generalize to many other crops and geographical regions remains uncertain.

3. Computational Requirements

DenseNet201Plus is more computationally demanding than lightweight models because additional attention and dense components are included.

4. Training Time

The additional architectural components slightly increased training time compared with the base model.

5. Possible Overfitting

Very high accuracy, particularly **99.50%** on the Black Gram dataset, should be interpreted carefully because the test dataset is relatively small.

6. Real-World Validation

Performance on controlled/public datasets does not necessarily guarantee equivalent performance under all field conditions such as different lighting, camera quality, backgrounds and disease severity.

12. Critical Analysis

Is DenseNet201Plus Appropriate?

Yes. DenseNet is appropriate because plant disease recognition depends heavily on detailed visual patterns such as spots, lesions, texture and color changes. Dense connections allow these features to be reused efficiently.

Is the Dataset Adequate?

The datasets are adequate for an experimental study, but they are **not fully sufficient for claiming universal real-world generalization**. More images from different geographical locations, crops, cameras and environmental conditions would strengthen the study.

Are the Evaluation Metrics Appropriate?

Yes. Accuracy, precision, recall and AUC are appropriate for classification. However, additional metrics such as **F1-score, balanced accuracy and per-class sensitivity** would provide a more complete analysis, particularly if class distributions are unequal.

Are the Results Convincing?

The results are promising because the proposed architecture consistently outperformed the compared models. However, stronger evidence could be obtained through:

- Multiple random train/test splits
- Cross-validation
- Statistical significance testing
- Independent external datasets
- Field testing

Could Another Unit IV Technique Perform Better?

Possibly. A **Vision Transformer (ViT)** or convolutional-transformer hybrid could potentially capture global relationships in complex leaf images. Lightweight architectures such as **MobileNet** could be preferable for smartphone deployment because they require fewer computational resources.

Therefore, DenseNet201Plus is a strong research solution, but it should be compared against newer architectures before large-scale deployment.

13. Future Scope

The authors suggest further improvements in preventing underfitting and overfitting and recommend deeper comparisons with newer architectures such as convolutional transformers and Vision Transformers.

Additional future directions include:

1. Collect larger and more diverse field datasets.
2. Include more crops and disease categories.
3. Perform cross-validation and external validation.
4. Compare DenseNet201Plus with Vision Transformers.
5. Develop lightweight versions for smartphones and edge devices.
6. Optimize inference speed and memory requirements.
7. Use advanced augmentation and synthetic image generation.
8. Improve explainability using additional XAI techniques.
9. Test the model under different lighting and environmental conditions.
10. Develop a real-time mobile application for farmers.

14. Conclusion

The reviewed research article presents **DenseNet201Plus**, an enhanced transfer-learning architecture for plant leaf disease identification. The researchers modified DenseNet201 by

incorporating improved preprocessing and augmentation, dense blocks, attention mechanisms, attentive transitions, batch normalization, global average pooling and improved dense layers. The model was evaluated on **Banana Leaf Disease** and **Black Gram Leaf Disease** datasets containing a combined **2,296 images**. It achieved **90.12% accuracy on the Banana dataset and 99.50% accuracy on the Black Gram dataset**, outperforming the evaluated baseline CNN architectures.

Overall, the article provides a **strong and relevant application of DenseNet and Transfer Learning** to agriculture. Its major strengths are feature reuse, high classification performance and improved interpretability through Grad-CAM++. However, the relatively small datasets and limited external validation mean that further field-based testing is required before claiming broad real-world generalization.

Overall assessment: Level 4 – Exemplary, because the study demonstrates a relevant real-world problem, an advanced deep-learning methodology, quantitative comparison, practical application and meaningful future research directions.

15. References

1. Mazumder, M. K. A., Kabir, M. M., Rahman, A., Abdullah-Al-Jubair, M., & Mridha, M. F. (2024). *DenseNet201Plus: Cost-effective transfer-learning architecture for rapid leaf disease identification with attention mechanisms*. **Heliyon**, **10(15)**, e35625. DOI: 10.1016/j.heliyon.2024.e35625.
2. Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). *Densely Connected Convolutional Networks*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
3. Pan, S. J., & Yang, Q. (2010). *A Survey on Transfer Learning*. **IEEE Transactions on Knowledge and Data Engineering**, **22(10)**, 1345–1359.
4. Selvaraju, R. R., et al. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization*. Proceedings of the IEEE International Conference on Computer Vision.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis